

Lab: Decision-Making and Prefrontal Function

Learning Goal: Analyze neural decision-making through Active Inference, focusing on EFE decomposition and dopaminergic control.

Part 1: Prefrontal Policy Evaluation

Exercise: For a complex decision (choosing a career path), map each PFC region's contribution:

PFC Region	Function	What It Computes for Career Choice
dIPFC	Working memory / policy maintenance	Holds multiple career options in mind simultaneously
vmPFC	Pragmatic value computation	Estimates expected reward (salary, satisfaction) for each option
ACC	Conflict monitoring	Signals when options are close in value (high decision difficulty)
Frontopolar (BA10)	Counterfactual reasoning	Evaluates the "road not taken" — what if I chose differently?
OFC	Subjective value	Integrates personal preferences with objective outcomes

Now trace a complete decision sequence: initial consideration → evaluation → conflict → resolution → commitment. What neural signal dominates at each stage?

Write your response here

Part 2: EFE Decomposition in Paradigms

Learning Goal: Identify pragmatic and epistemic value in experimental paradigms.

Exercise: For each paradigm, identify the pragmatic and epistemic components:

Paradigm	Pragmatic Value (reward)	Epistemic Value (information)	Which Dominates?
Two-armed bandit (exploitation phase)	Choosing the known high-reward arm	None — uncertainty already resolved	Pragmatic
Two-armed bandit (exploration phase)	Unknown reward at new arm	Sampling the new arm reveals its value	Epistemic
Foraging (rich environment)	Staying at current patch	Moving to unknown patch to estimate quality	Trade-off
Medical diagnosis	Prescribing treatment	Running diagnostic tests	Epistemic first, then pragmatic
Scientific experiment	Applying existing knowledge	Running experiment to reduce uncertainty	Epistemic

For the foraging scenario, explain how the exploration-exploitation trade-off is resolved by EFE.

Write your response here

Part 3: Dopamine and Decision Confidence

Learning Goal: Analyze how dopaminergic dysfunction affects decision-making.

Exercise: Complete the clinical analysis:

Condition	Dopamine Status	Policy Precision (γ)	Behavioral Consequence	Active Inference Explanation
Healthy	Normal tonic DA	Moderate γ	Appropriate balance of decisive and flexible behavior	Precision-weighted EFE evaluation
Parkinson's (early)	Depleted tonic DA	Low γ	Indecisiveness, bradykinesia, apathy	Flat policy distribution → no clear winner
Schizophrenia (acute)	Elevated DA	High γ	Jumping to conclusions, aberrant salience, delusions	Overconfident belief in a single policy/model
ADHD	Dysregulated DA	Variable γ	Impulsive switching, difficulty maintaining focus	Moment-to-moment precision fluctuations
L-DOPA medication	Artificially elevated	Excessively high γ	Impulsive gambling, hypersexuality	Overshooting precision → compulsive exploitation

For each condition, explain how medication or therapy could aim to recalibrate γ .

Write your response here

Part 4: Working Memory Experiment Design

Learning Goal: Design an experiment testing the Active Inference account of working memory.

Exercise: Design an experiment to test the hypothesis that working memory capacity is limited by the number of gamma bursts within a theta cycle:

1. **Participants:** How many? What exclusion criteria?
2. **Task:** What working memory task? (e.g., sequential recall, change detection, spatial working memory)
3. **Neural measurements:** What brain signals would you record? (EEG/MEG for theta-gamma coupling)
4. **Predictions:** If the theta-gamma coupling hypothesis is correct, what specific result would you expect as working memory load increases from 1 to 6 items?
5. **Alternative hypothesis:** What would the result look like if working memory is not oscillation-based?

Write your response here

Part 5: Reflection

In 300 words, reflect: The neural decomposition of EFE into pragmatic and epistemic components is elegant, but real-world decisions rarely have clean pragmatic/epistemic separation. How well does this decomposition handle genuinely ambiguous decisions (e.g., choosing whether to have children, deciding whether to change careers)?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Neural mapping	PFC policy evaluation
2	Paradigm analysis	EFE decomposition
3	Clinical reasoning	Dopamine and decision confidence
4	Experiment design	Working memory oscillations
5	Critical reflection	Limits of decomposition